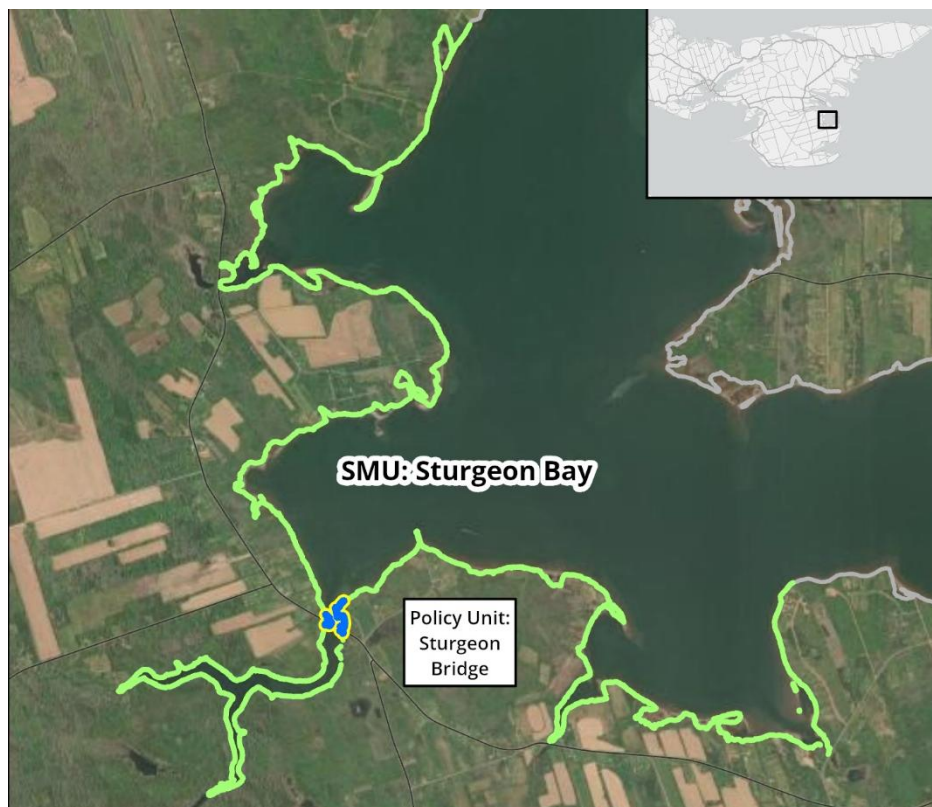


24. Sturgeon Bay

Shoreline length: 34 km



Shore Types

This SMU includes Clarks Bay and Sturgeon Bay, sheltered behind Panmure Island. Main shore types are low plains and wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
71	3	0	0	26	0	0	0



Thornton Point and Clarks Bay inside Sturgeon Bay

Governance, Infrastructure, and Economy

Sturgeon Bay falls almost entirely within the Rural Municipality of Three Rivers; approximately 10% of the shore of the estuary is unincorporated.

Key infrastructure includes the Sturgeon wharf and Old Wharf Road, Route 17, Pembroke Rd and St. Andrews Point Road. Coastal economic drivers include Sturgeon Wharf which supports an active fishery.

Local and legacy land use constraints include several subdivisions with small lots on unpaved roads running parallel or perpendicular to the coast.

Planning and Policy Framework

Where provincial rules apply, land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a

coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted. Within Three Rivers, the Town's municipal rules apply. Development on the coast is informed by the Environmental Protection Act, including a 15m buffer, with no additional building setbacks aside from the buffer. The Official Plan includes specific guidance permitting armouring, though such provisions are secondary to any provincial standards, and includes some policies regarding expansion of buildings within the sea level rise overlay. There are no specific flood elevations or vertical setbacks established, but the Official Plan does include specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Sturgeon Bay falls entirely under municipal planning authority of Three Rivers. Coastal development is informed by provincial legislation, including the Environmental Protection Act. The Three Rivers Official Plan includes policies regarding expansion of buildings in areas affected by sea level rise. No specific flood elevations or vertical setbacks are established.

Official plan includes specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Economic drivers include Sturgeon Wharf which supports an active fishery. Local land use constraints include several subdivisions with small lots on unpaved roads running parallel or perpendicular to the coast.

Sediment transport

The estimated net sediment transport rate averaged across the SMU is moderate. It feeds a few beaches extending into the bay such as Thornton Point, Meadowfoot Point and Wight Point.

Coastal Flood and Erosion Risk

While low plains and wetland shores are vulnerable to increasing flood hazards, infrastructure risk is low. Erosion hazard is limited due to the sheltered nature of the area except along the more exposed headlands. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline]

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	6 [<1/km]	8 [<1/km]	11 [<1/km]	13 [<1/km]	0	4 [<1/km]	14 [<1/km]
Roads	1278 m [38 m/km]	1449 m [43 m/km]	1716 m [51 m/km]	2664 m [79 m/km]	46 m [1 m/km]	107 m [3 m/km]	377 m [11 m/km]

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
24.00 Sturgeon Bay	NAI	MR	MR
Rationale	Given the low risk levels, the initial strategy is No Active Intervention. Mid- to long-term MR will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including wetlands and low plains with increasing flood hazard.		
24.01 Sturgeon Bridge -PU	R	R	R
Rationale	Resist strategy would be suitable for bridge, harbour and adjacent shores, but will need to incorporate flood mitigation.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.